Email Bounce Rate Analyzer

SMTP Data Investigation — Root Cause Analysis Project

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1. Project Overview

This project analyzes 6 months of SMTP email response data (February to August 2018) to investigate a reported increase in email bounce rates. The goal is to identify the root cause using data analysis techniques and propose actionable solutions.

This is an independent data analysis project demonstrating skills in Python, SMTP protocol understanding, root cause analysis methodology, and data visualization.

2. Investigation Methodology

The investigation followed a structured analytical approach:

- Validate the reported increase in bounce events across the dataset
- Compare SMTP response trends across all months in the dataset
- Identify the SMTP error code contributing most to the increase
- Analyze affected email domains to pinpoint the source
- Develop a root cause hypothesis supported by data evidence

3. Investigation Steps

Step 1 — Understanding The Data

- Dataset contains 167 rows of SMTP response data across 6 months
- Column 1 = UNIX timestamp converted to readable dates
- Remaining columns = SMTP response codes per email domain
- Key domains include gmail.com, hotmail.com, centrum.sk, seznam.cz, yahoo.com and others
- SMTP code 200 = successful delivery, all other codes = bounce or error

Step 2 — Data Preparation

- UNIX timestamps were converted into readable dates using Python pandas library
- Data was grouped by month to compare bounce rates across the 6 month period
- Total successful deliveries and total bounces were calculated per month
- Bounce rate percentage = total bounces divided by total emails times 100

Step 3 — Monthly Bounce Trend Analysis

Monthly SMTP bounce events were compared across the six-month period.

May 2018 showed a significant increase in bounce events compared with previous months. Additional elevated bounce activity continued into June before gradually decreasing in July and August.

This established May 2018 as the starting point for detailed investigation.

Step 4 — Error Code Analysis

Bounce codes in May 2018 were analyzed in detail:

- Code 421 — Server temporarily unavailable: 204 occurrences
- Code 451 — Server error try again: 210 occurrences
- Code 452 — Insufficient storage gmail.com: 721 occurrences
- Code 499 — Connection timeout: 266 occurrences
- Code 550 — Invalid email address hard bounce: 636 occurrences
- Code 552 — Mailbox full: 450 occurrences
- **Code 554 — Transaction failed: 2,835 occurrences — DOMINANT ERROR**
- Code 605 — Unknown delivery error: 639 occurrences

Code 554 accounts for the largest share of bounces in May 2018 — significantly higher than any other error code.

Step 5 — Domain Level Analysis

Breaking down error code 554 by email domain:

- centrum.sk — 1,884 occurrences of error 554 in May 2018
- Other domains — 884 occurrences combined
- centrum.sk contributed 66% of all 554 errors in May

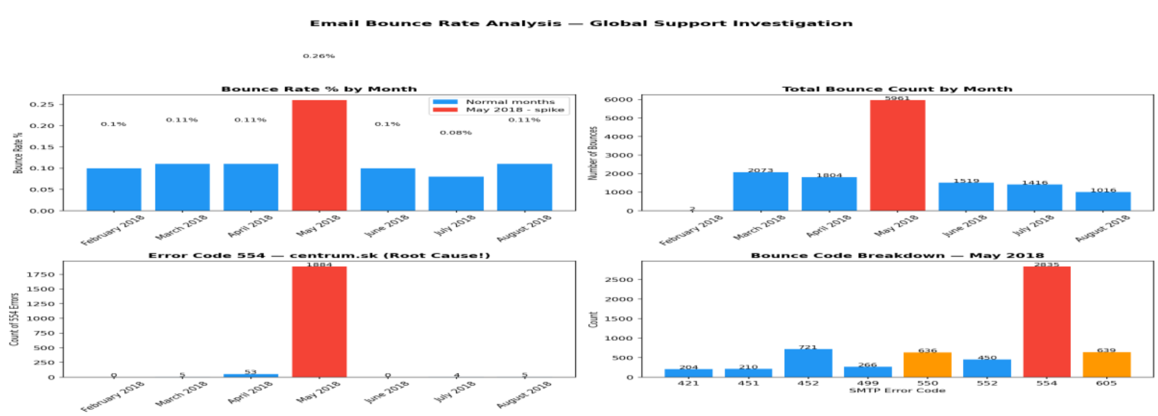
centrum.sk error 554 trend across months:

- February 2018: 0
- March 2018: 5
- April 2018: 53
- **May 2018: 1,884 — massive spike**
- June 2018: 0
- July 2018: 4

The spike is isolated to May 2018 and almost entirely concentrated on centrum.sk. This is not a gradual trend but a sudden sharp increase that reversed completely in June.

4. Data Visualization

The following charts summarize the key findings from the investigation:



Charts show: Monthly bounce trend confirming May spike, SMTP error code breakdown showing code 554 as dominant error, and domain analysis confirming centrum.sk as primary affected domain.

5. Root Cause Analysis

Based on the data analysis the most likely root cause is:

centrum.sk email domain experienced a temporary server-level policy or configuration change in May 2018 that caused it to reject email deliveries with SMTP error 554 (Transaction Failed). This is supported by the following evidence:

- Error 554 on centrum.sk spiked from 53 in April to 1,884 in May — a 35x increase
- The spike is isolated only to centrum.sk and only to May 2018
- The error completely disappeared in June suggesting a temporary configuration issue
- Error 554 indicates the receiving server rejected emails during SMTP transaction — a recipient-side issue
- Other domains show normal bounce rates in May confirming the issue is domain specific

Secondary contributing factors in May:

- Code 452 on gmail.com increased to 721 — storage or rate limiting issues
- Code 550 increased to 636 — invalid email addresses in the list
- Code 605 increased to 639 — unknown delivery errors on multiple domains

6. Recommended Actions

Immediate Actions

- Contact centrum.sk postmaster to understand why 554 errors spiked in May
- Check if the sending domain or IP was temporarily blacklisted by centrum.sk
- Review email sending practices for compliance with centrum.sk policies
- Verify if centrum.sk made server configuration changes in May 2018

Email List Hygiene

- Clean email list to remove invalid addresses causing 550 hard bounce errors
- Implement email validation before adding new addresses to mailing list
- Separate centrum.sk addresses and test delivery before resuming campaigns

Monitoring and Prevention

- Set up real-time SMTP error monitoring to catch spikes early
- Create alerts when bounce rate exceeds 2% for any domain
- Implement domain-level bounce rate tracking as standard practice
- Consider email deliverability tools like Mailgun or SendGrid for automated bounce management

7. Assumptions

- The dataset accurately represents SMTP response activity for the period analyzed
- Campaign volume or total email sent data was not available in the dataset
- Findings are based solely on SMTP response trends contained in the dataset

8. Summary

This investigation identified a significant increase in SMTP bounce events during May 2018. Analysis of SMTP response codes showed that error 554 (Transaction Failed) was the dominant contributor. Further domain-level analysis revealed that the majority of these errors originated from centrum.sk, where error 554 increased dramatically in May before returning to normal levels in June.

The most likely root cause is a temporary recipient-side policy or configuration issue at the centrum.sk domain. Recommended next steps include contacting the domain administrator, reviewing sender reputation and authentication, validating mailing lists, and implementing domain-level SMTP monitoring.

9. Tools and Technologies Used

- Python 3 — pandas, matplotlib, seaborn
- Data Analysis — SMTP response data parsing and aggregation
- Data Visualization — matplotlib charts
- Root Cause Analysis — structured investigation methodology
- SMTP Protocol Knowledge — error code interpretation

— End of Report —

github.com/BidyutRajkhowa/email-bounce-analyzer